UUID:D3BBEAC7-490C-4CF3-AB2A-966DB9531A41
TCGA-BJ-A0ZH-01A-PR Redacted

PATIENT HISTORY:
CHIEF COMPLAINT/PRE-OP/POST-OP DIAGNOSIS: Thyroid nodule.
PROCEDURE: Not answered.
SPECIFIC CLINICAL QUESTION: Not answered.
OUTSIDE TISSUE DIAGNOSIS: Not answered.
PRIOR MALIGNANCY: Not answered.
CHEMORADIATION: Not answered.
ORGAN TRANSPLANT: Not answered.
IMMUNOSUPPRESSION: Not answered.
OTHER DISEASES: Not answered.

ICD-0-3
carcinoma, papillary, tall cell variant 8344/3
Site: thyroid, NOS C73.9 lw 11/27/11

ADDENDA:
Addendum

MOLECULAR ANATOMIC PATHOLOGY TESTING:

A. ***BRAF* V600E mutation IDENTIFIED.**
B. **Mutations in *NRAS61*, *HRAS61*, *KRAS12/13*, and *RET/PTC1*, *RET/PTC3* and *PAX8/PPARg* rearrangements NOT identified.**

NOTE:
Nucleic acids were extracted in the amount sufficient for testing. Amplification of *GAPDH* and *KRT7* controls was acceptable.

BACKGROUND:
Mutations in either *BRAF* or *RAS* genes or *RET/PTC* rearrangements are found in more than 70% of papillary thyroid carcinomas (1). *BRAF* V600E (T1799A) mutation has been associated with more aggressive behavior of papillary carcinoma (2) and may be used for preoperative risk stratification of patients with papillary thyroid cancer (3). Mutations in the *RAS* genes or *PAX8/PPARg* rearrangement occur in ~70% of follicular thyroid carcinomas and with lower frequency in oncocytic (Hürthle cell) carcinomas (4). Based on a recent prospective study of FNA samples (5) and our experience at detection of *BRAF* mutations and *RET/PTC* and *PAX8/PPARg* rearrangements in thyroid FNA samples correlated with ~100% probability of cancer, whereas detection of *RAS* mutation correlated with malignancy in 83-87% of cases and with benign follicular adenoma or hyperplastic nodule in 13-17%. However, single cases of false-positive *BRAF* mutation detection in thyroid FNA samples have been reported in the literature (6) and therefore the results of molecular testing should be interpreted in combination with clinical information, imaging, and cytology.

1. Adeniran AJ, et al. *Correlation between genetic alterations and microscopic features, clinical manifestations, and prognostic characteristics of thyroid papillary carcinomas. Am J Surg Pathol 2006, 30:216-222*
2. Elisei R, et al. *BRAF V600E mutation and outcome of patients with papillary thyroid carcinoma: a 15-year median follow-up study. J Clin Endocrinol Metab 2008, 93:3943-3949.*
3. Xing M, et al. *BRAF mutation testing of thyroid fine-needle aspitation biopsy specimens for preoperative risk stratification in papillary thyroid cancer. J Clin Onclology 2010, 27:2977-82.*
4. Nikiforova MN, et al. *RAS point mutations and PAX8-PPARγ rearrangement in thyroid tumors: Evidence for distinct molecular pathways in thyroid follicular carcinoma. J Clin Endocrinol Metab 2003, 88: 2318-26.*
5. Nikiforov YE, et al. *Molecular Testing for Mutations in Improving the Fine Needle Aspiration Diagnosis of Thyroid Nodules. J Clin Endocrinol Metab. 2009, 94(6):2092-8.*
6. Chung KW, et al. *Detection of BRAFV600E mutation on fine needle aspiration specimens of thyroid nodule refines cyto-pathology diagnosis, especially in BRAF600E mutation-prevalent area. Clin Endocrinol (Oxf). 2006, 65(5):660-6.*

MUTATIONAL ANALYSIS:
Thyroid FNA samples were collected in the nucleic acid preservative solution. For cytology samples and frozen samples, extraction of DNA and RNA was performed from the sample provided using standard laboratory procedure. Optical density readings were obtained. Real-time PCR was performed on the platform to amplify BRAF codons 599-601, NRAS codon 61, HRAS codon 61, and KRAS codons 12/13 sequences. Post-PCR melting curve analysis was used to detect possible mutations. Single-step real-time RT-PCR was performed to amplify the fusion points of RET/PTC1, RET/PTC3, and PAX8/PPARg rearrangements on ABI7500. If required, the mutation type was confirmed by Sanger sequencing of the PCR products on Amplification of the GAPDH and KRT7 genes was used to control quality and quantity of RNA and amount of epithelial cells present in the specimen. DNA or RNA from samples positive for each of these mutations were used as positive controls. Amplification at 35 cycles or earlier was considered sufficient for the analysis. The limit of detection is approximately 10-20% of alleles with mutation present in the background of normal DNA and RNA.

FINAL DIAGNOSIS:

PART 1: THYROID GLAND, TOTAL THYROIDECTOMY (12.7 GRAMS) –
A. **PAPILLARY THYROID CARCINOMA, TALL CELL VARIANT, 2.1 CM, LOCATED IN LEFT LOBE.**
B. **NO EXTRATHYROIDAL EXTENSION.**
C. **ALL MARGINS ARE NEGATIVE.**
D. **ONE BENIGN LYMPH NODE (0/1).**
E. **ONE NORMOCELLULAR PARATHYROID GLAND.**
F. **AJCC TUMOR STAGE pT2 N1b.**

Criteria	Yes	No
Diagnosis Discrepancy		X
Primary Tumor Site Discrepancy		X
HIPAA Discrepancy		X
Prior Malignancy History		X
Dual/Synchronous Primary Noted		X
Case is (circle): QUALIFIED / DISQUALIFIED		
Reviewer Initials MG Date Reviewed: 11/27/11		

lw 11/27/11

PART 2: LYMPH NODE, CENTRAL COMPARTMENT, EXCISION –
ONE OUT OF FOUR LYMPH NODES POSITIVE FOR METASTATIC CARCINOMA (1/4).

[redacted]

COMMENT:
Molecular studies will be performed on frozen banked tissue form the papillary carcinoma and results will be reported as an addendum.

CASE SYNOPSIS:

SYNOPTIC DATA - PRIMARY THYROID TUMORS

SPECIMEN TYPE:	Total Thyroidectomy
TUMOR SITE:	Left Lobe
TUMOR FOCALITY:	Unifocal
TUMOR SIZE (largest nodule):	Greatest Dimension: 2.1 cm
HISTOLOGIC TYPE:**	Papillary carcinoma
PRIMARY TUMOR (pT):	pT2
REGIONAL LYMPH NODES (pN):	pN1a Number of regional lymph nodes examined: 5 Number of regional lymph nodes involved: 1
EXTRANODAL EXTENSION:	Not identified
DISTANT METASTASIS (pM):	Not applicable
EXTRATHYROIDAL EXTENSION:	Not identified
MARGINS:	Margins uninvolved by carcinoma
LYMPH-VASCULAR INVASION:	Not identified
